Optimizing Grantshire's Gritting Service: A Comprehensive Analysis of Grantshire's Gritting Depot Development and Relocation Options

Executive Summary

In this report, we present a thorough analysis of options for the development or relocation of Grantshire Council's gritting depots. In our analysis we focus on efficiency and cost-effectiveness, while keeping public safety as a main priority. After evaluating logistical considerations, accident rates on different road types and operational costs under varying weather conditions, our ultimate recommendation is option 6b. We have determined depot Z to be the best option for its robust responsiveness to harsh weather conditions, while minimising environmental impact and maximising resource utilisation. This is due to its central positioning, which provides the most cost-effective solution to the problem of putting salt on the roads through limiting the dead mileage on all the routes. Additionally, a brand-new depot encourages investment in technology and infrastructure to improve fleet longevity, which will further enhance Grantshire's ability to ensure safe road conditions for the foreseeable future. By considering our recommendations, Grantshire can effectively safeguard the wellbeing of its' citizens, and optimise road maintenance operations throughout this investment life cycle.

Contents

1.1 Introduction

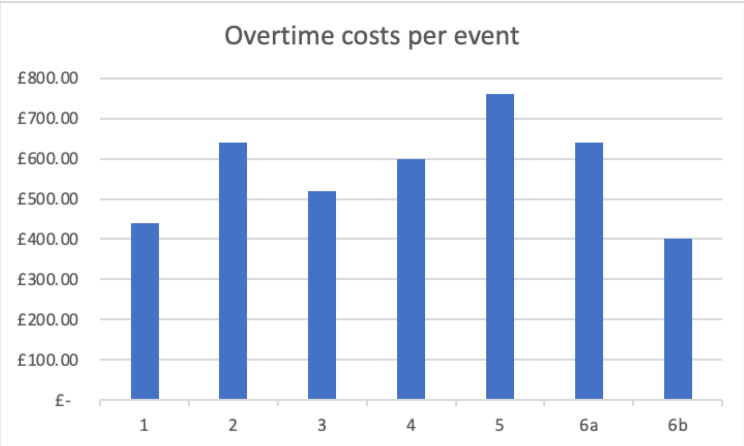
This report outlines the research conducted on the possible development and/or relocation of Grantshire Council grit salt depots, essential for servicing the surrounding areas during various winter conditions. Our analysis encompasses a range of possible weather conditions from ‘extreme’ to ‘mild’, ensuring the analysis can provide comprehensive insights for all possibilities.

We delved into the cost and logistical implications of each option, with a focus on optimising the council’s operations. Furthermore, all considerations align with the key performance indicators established by the council, both emphasising value for money and ensuring that safety is a key priority.

2. Depot options analysis

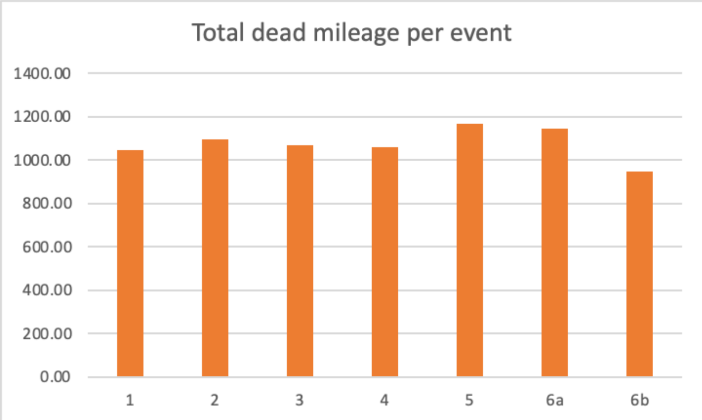
2.1 Analysis of human and environmental logistical factors

One of the key considerations is the human and environmental logistical factors that a possible development and or relocation would require. For example, the amount of time each employee works must be considered to ensure that they aren't overworked to maintain a high level of employee satisfaction and wellbeing. Furthermore, an increased amount of employee overtime will also increase costs to the council in the form of extra wages so a well spread workload between the 16 drivers would be optimal. As can be seen below options 1 and 6b performs the best with this objective in mind at an extra cost of £440 and £400 per event respectively. Option 5 performs much worse with an overtime cost of £760 per event. The other options are relatively similar at around £600 per event and can also be seen in Figure 1.



(Figure 1: showing overtime costs per event for each option)

The impact which the dead mileage has on the environment must also be a consideration, because increased wasted mileage will lead to a bigger impact on the environment as well as higher wasted fuel costs. The dead mileage also contributes to the previously mentioned overtime, as workers take longer to get to the start of their routes, leading to increased shift times. The wasted time and productivity, as well as the increased emissions we believe make this a key consideration. Once again 6b performs considerably better than some of the other options, with a total dead mileage per event of 946. The other options perform similarly at around 1100 dead mileage per event. This 150 mile difference can add up significantly across a year, for example in an year that experiences an average amount of events the difference in total dead mileage would be 6300 miles. In an extreme winter the difference is even larger (9000 miles), these values reinforce the importance of considering the dead mileage which each option offers.



(Figure 2: showing dead mileage per event for each option)

Distance to start must also be considered, as if the depot is centrally located then it is likely to reduce the response time to adverse weather conditions. This would mean faster action for the council, which would increase the level of safety provided and potentially save lives. While saving lives is of paramount importance, it would also save the council money because as suggested by a report conducted by the Official Department for Transport Accident Costs, it was found that a fatal injury would cost the very high figure of £1,447,490 per accident. A serious injury would also be very costly for the council at £168,260, so it is clear to see the need for any improvements which can help avoid these types of accidents to be sought. With this factor in mind, option 3 performs the best as it has an average distance to start of 13.75 miles- considerably smaller than the worst option, option 6a, at 23.2 miles. Across 16 routes this equates to a difference of 150 miles per event, which clearly shows the difference in distances to each start and the potential impact it could have.

3. Gritting strategy

3.1 Prioritisation by Road Type

In Grantshire, around half (52%) of roads are currently gritted. The National Audit Office recommends a minimum of 35%, and across England the overall average is 38% with a high of 75%. This percentage is more than acceptable from this perspective, as Grantshire exceeds the recommended minimum as well as the national average. This extensive coverage likely contributes to lower accident rates compared to other regions, ultimately helping keep Grantshire’s roads safer than the rest of the nation on average.

Road Type	Fatal	Serious	Slight	BVKM
Highways Agency	3.2	16.8	142.9	150
Principal A	10.9	87.9	520.6	112

Minor (Urban)	6.7	85.2	610.3	105
Minor (Rural)	9.5	94.5	420	38

(Figure 3: Accidents by Type of Road)

According to relative accident rates in terms of traffic volumes (measure in billion vehicle kilometres, BVKM) in England in 2019 (Figure 3), the most dangerous roads for serious and fatal accidents are rural minor roads and principal A roads, therefore these are a priority for gritting. In England as a whole, around 88.65% of rural minor roads are unsalted, compared to 55.87% of urban minor roads. Because of this, we would recommend targeting more rural roads as there is significant room for improvement, especially considering how dangerous these roads are.

The spreading rate is set to 15g/m^2, which is above the recommended minimum of 10g/m^2. This suggests an effort to maximise effectiveness of each event, as some material gets wasted by getting blown away off the road, for example. If it was necessary, for example to cut costs, then there is space to decrease the spreading rate so that it would still be within the recommended amounts. There is nothing to suggest that increasing the spreading rate would add any benefit, apart from simply being more costly. There is an argument to be made that less salt should be used, as it causes corrosion to metallic infrastructure, concrete and vehicles. However, the costs of these are minimal in comparison to the financial and social costs of accidents caused by not gritting roads enough.

3.2 Gritting depot capacity and costs

We also broke down the numbers on the amount of grit needed across each type of weather conditions. For an average winter we would use just over 6,000 tonnes of grit- all options are able to hold this amount therefore none will be ruled out. However, for a harsh winter we estimated to use just over 8600 tonnes of grit. This exceeds the capacity of any of the depots. Therefore, we would need to factor in how easy it is to reorder grit and how accessible it is to deliver to the depots. Using two depots may be inferior to using just one, as in a harsh winter we will need to restock grit and may be on short notice. Delivering to two different depots may cause problems with distributing the material across the trucks in each depot. However, delivering to one depot may be more convenient as all the grit will be stored in one place and be accessible for all the trucks.

During an average winter season, the Authority uses around 6,000 tonnes of grit salt for precautionary gritting. The depot capacity is 7,000 tonnes, which leaves a buffer of around 1,000 tonnes on average. Last season 5,463 tonnes were used over 38 events, therefore the stockpile was more than enough. However, last season was relatively mild so this may not be the case every year. If we had another winter like in 1967/68 which had the highest number of events since 1965 with 68 events, we would need nearly 10,000 tonnes of grit salt, which could cost over £250,000 on salt alone. This would go way over the stockpile capacity and force us to order at least 2/3,000 extra tonnes at the higher price of £35 a tonne. This also bears the practical risks of shipment delays due to extreme weather conditions, or the small chance of re-stocking becoming problematic. This has never happened yet, and 1967/68 was an extreme example. However, these sorts of extreme weather events may become more frequent due to climate change, so increasing the stockpile capacity may be worth considering if weather trends suggest more frequent harsh winters in the future.

Looking further at the costs for grit, on a low end a tonne of grit would cost £25 and on the high end it would cost £35. From our grit values for a harsh and average winter we were able to determine that the low-end cost for grit in a harsh winter is £215,640 and for high end cost it would be £301,896, which is a 40% difference in cost. This is a substantial jump in price- we can infer over a long period of time the additional cost of purchasing grit for £35 would be significant. Over a 30-year period we see the difference of costs rise to just over £2,500,000. Therefore, in order to have a cost-efficient model for the whole operation we would need to minimise our operational cost to balance out the cost of grit if it was bought at the high end. Using the most cost-efficient option to tackle the gritting problem could allow investment elsewhere.

4. Gritter availability and operational effectiveness

Investing in upgrading and maintaining our fleet of gritting trucks is essential in ensuring operational effectiveness during winter seasons. With this investment, we can enhance the quality of maintenance and overall availability of gritters, while optimising their performance to meet the demands of varying weather conditions. Focusing on the longevity of this investment allows us to take advantage of technological advancements and incorporate these into options such as 6B, where we can tailor the depot from the ground up to ensure all of Grantshire's needs can be met over the long term.

Addressing breakdowns and maintenance issues promptly is crucial for minimising disruptions to our gritting operations. Centralising our resources by establishing a new depot in location Z streamlines our response to breakdowns. By consolidating spare gritters and maintenance equipment in a single location, we can improve our fleet's responsiveness, mobility and longevity. This centralised approach enhances our ability to quickly address breakdowns, ensuring that our gritting trucks remain operational and ready to tackle winter weather conditions effectively.

As we make this long-term investment in our gritting infrastructure, it's essential to maintain flexibility in our capacity for grit salt and gritters. Regardless of the chosen depot location, we must keep the option of purchasing more gritters open, in order to accommodate potential expansion and evolving operational needs. By maintaining flexibility in our capacity for grit salt and gritters, we can adapt to changing demands and ensure that our gritting operations remain robust and effective over time. This approach allows us to future-proof our infrastructure and optimise our ability to respond to all types of winter weather conditions effecting Grantshire.

5. Cost analysis

5.1 Initial cost analysis

The potential cost of each option is a key consideration, especially given the council is looking to ensure any investment it makes has sufficient returns. After receiving the closing, running and initial costs for all the options, we found option 6a to have the lowest running costs at £20,000, beating the competition by 24%. This suggests that this option would be the cheapest in the long term.

However, we also had to factor in the initial and closing costs which saw an overall cost to start up being £268,000 for this option. This value calculated is on the high end of the scale when compared with other options, due to the fact that we would need to close current depots and open a brand new one. Similarly, option 6b also has a low running cost of £25,000 and overall start-up cost of £318,000. This analysis makes us confident that either of these two options would be the best long-term investment if the council has a heavy focus on reducing recurring monthly expenses.

We wanted to look at the operational costs under different eventualities to see the most cost-efficient model, as over a long period of time there is a high chance of a potential change in weather conditions. This has especially been the case recently with the true extent of climate change being realised. To factor these changing weather conditions into our analysis we did three sets of different calculations for our operational cost.

5.2 Cost analysis under differing weather conditions

First, we wanted to find what we could consider an average winter by looking at past data of how many events would normally occur each year. We calculated an average over several different time periods so we could gather a good estimate of occurrences. Additionally we calculated averages over 12 and 18 years, to capture possible trends that didn't occur in recent times, and such a long average would filter out outliers which will give us a more suitable and representative value for our events. We found the average values were between 41 and 42 and therefore took an average winter to be 42 events to be conservative.

From this we could calculate the operational cost for an average winter using our costing model. We also considered additional driver costs in the calculations to give us a complete overall operational cost. After adding the additional drivers' costs, we see 6b and 1 become the two most cost-efficient options at different points in time. Breaking it down to better understand this performance, these two options have the lowest additional driver cost, indicating that when prolonging the drivers journey by adding additional mileage on to the route, we don't benefit financially. Over a short period of time, we see option 1 being the most cost efficient with option 2 being in close second with just over a £30,000 difference over 5 years, this is due to the low start up and closing costs. After 20 years we see the low running and operational costs of our new depots becoming influential as option 6b and 6b close the overall cost gap. After 30 years we see option 6b and 1 only separated by a few hundred thousand pounds. This most cost-efficient solution over a 30-year period was 6b as costs came to £13,815,563. As expected, due to their low running costs one of the new depots became the most cost-efficient option over a longer period. However a factor not considered was the repair of equipment which is more likely to fail in the older depots which are expanding, as our new sites will have the new technology and the machines are more likely to be durable.

Secondly, we calculated costs for a mild winter to understand the operations under a best-case scenario. A value of 35 was used as this was the lowest number of events for a single year on record. Our findings were similar to the previously stated average winter conditions, where again we see, over a long period of time, our options for new depots becoming the most cost-efficient. Our calculations under a mild winter show option 1 being the cheapest over 30 years with a value of £9,902,000, with 6b close behind with only a £61,000 difference.

Finally, we calculated costs for a harsh winter, which was intended to show the worst scenario possible and how each option would perform under these conditions. This could be very useful for future-proofing the decision and ensuring the option picked could perform well under all eventualities. This is particularly important considering the context where, if underprepared for a harsh winter, the safety of the county's citizens could come under threat on icy roads. We suggested a harsh winter would contain 68 events, as this was the maximum events that occurred in a single year from across the past data. Once again, we saw similar results where over a short period of time option 1 and 2 are the most cost effective, whereas over a long period of time option 6b becomes superior and the most cost efficient. For a single year of 68 events the cheapest operational costs come to £476,142 which is option 6b, £7,800 cheaper than option 1 in second. Over 30 years including the initial costs we still have these two options being the cheapest with both being just over £14,600,000. However, it must be noted that this is the worst scenario possible and is therefore very unlikely to repeat consistently.

6. Conclusion

In conclusion, our analysis of the potential development and or relocation of the Grantshire Council grit salt depots provides useful insights into the optimisation of the winter operations which can allow the decision makers to make well informed decisions.

The consideration of human and environmental logistical factors details the importance of employee workload management and the minimisation of environmental impacts such as dead mileage. The option 6B performed particularly well in these areas, showcasing its potential to enhance operational efficiency while reducing environmental footprint.

Understanding the impact each type of road has on accident rates we found to be another key consideration, particularly rural minor roads, and principal A roads due to their higher accident rates. This highlights the importance of targeting gritting strategies to ensure the highest possible level of road safety whilst using the least possible resources. Despite Grantshire's gritting coverage surpassing national recommendations, there is a lot of space for improvement, especially in targeting rural areas where significant safety enhancements can be achieved.

Furthermore, the analysis of gritting depot capacity and costs reveals the necessity of balancing operational efficiency with financial considerations. Options like 6b emerge as cost-effective solutions, particularly under harsh winter scenarios, demonstrating the value of new investment in infrastructure and the improvements it can make to the level of efficiency.

The consideration of operational costs under differing weather conditions is key as its emphasis must be put on future-proofing decisions giving the level of investment and ensuring cost-efficiency over the long term. By integrating new technological advancements, and improving capacity planning, Grantshire can ensure its gritting operations remain robust and effective in mitigating the impacts of winter weather conditions long into the future.

Considering these findings, the final recommendations prioritise options that offer the best balance between operational efficiency, cost-effectiveness, and safety considerations which are key to the council. By adopting a comprehensive approach that addresses both immediate needs and long-term sustainability, Grantshire can optimise its gritting operations to better serve its community and ensure safer road conditions during winter months.

7. Final recommendations

After analysing and evaluating all our options concerning the development or relocation of Grantshire’s grit salt depots, option 6B emerged as the optimal choice. Despite initial startup costs, its long-term operational efficiency outweighs these expenses, in large part due to its central position; ensuring minimal dead-mileage, rapid responsiveness to breakdowns and resilience to different weather scenarios including the more severe winters Grantshire may face in the future. This reliability while tackling a harsh winter while also having the cheapest operational cost of any of the options makes 6B the best value for money over time, ideal for a long-term investment such as this.

Adding to this, with 6B being a completely new depot, investment can be put into the technology built into it to further optimise everything from gritter maintenance to a potentially more suitable grit salt capacity that ensures Grantshire is stocked up and can avoid re-ordering where possible. Furthermore, the advantages of building a centralised gritting depot from the ground up stretch to the availability and longevity of our gritting fleet, leaving us the option to expand the fleet as well as having all spare parts and mechanics on one modern site, increasing the quality and speed of any repairs that may need to be carried out.

By adopting these recommendations and selecting Option 6b, Grantshire can ensure robust and reliable gritting operations, effectively safeguarding the well-being of its’ citizens.